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MARINE VESSEL STEERING CONTROL DEVICE AND METHOD, AND MARINE VESSEL

Abstract

A marine vessel steering control device includes a first imager fixed to a hull to capture an image of a subject, and a controller configured or programmed to include a vessel operation controller to perform an automated vessel operation of the hull based on a captured image obtained by the first imager, and a determiner to determine, based on an imaging angular field of the first imager, whether or not sunlight is located within a reference angular field defined as an angular field within the imaging angular field. In a case where the sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change an actual traveling direction of the hull to an angle at which the sunlight is located outside the reference angular field.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of Japanese Patent Application No. 2024-31010, filed on Mar. 1, 2024, which is hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] The present invention relates to marine vessel steering control devices and methods, and marine vessels.

2. Description of the Related Art

[0003] In the automotive field, automated driving technologies have been developed. Generally, images captured by cameras are used for automated driving. Since, in the captured images obtained in a state in which sunlight enters an imaging angular field of a camera, there is a possibility that an imaged subject or the like cannot be properly captured, it is desirable to avoid backlit conditions in which sunlight enters the imaging angular field.

[0004] In this respect, in Japanese Laid-open Patent Application (Kokai) No. 2022-155838, when a backlit condition is predicted, backlight avoidance processing such as a change in vehicle location, a detour, or a change in passing timing is executed.

[0005] However, since the marine vessel field has unique circumstances different from those on land, such as the absence of roads, taking the circumstances into consideration allows room for studies of approaches to navigation with appropriate avoidance of reflected sunlight.

SUMMARY OF THE INVENTION

[0006] Example embodiments of the present invention provide marine vessel steering control devices capable of reducing deterioration in quality of an acquired image due to sunlight and achieving appropriate automated vessel operations.

[0007] According to an example embodiment of the present invention, a marine vessel steering control device includes a first imager fixed to a hull to capture an image of a subject, and a controller configured or programmed to include a vessel operation controller configured or programmed to perform an automated vessel operation of the hull based on a captured image obtained by the first imager, and a determiner configured or programmed to determine, based on an imaging angular field of the first imager, whether or not sunlight is located within a reference angular field defined as an angular field within the imaging angular field, wherein, in a case where the sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change an actual traveling direction of the hull to an angle at which the sunlight is located outside the reference angular field.

[0008] According to this configuration, the determiner is configured or programmed to determine, based on the imaging angular field of the imager, whether or not sunlight is located within the reference angular field defined as an angular field within the imaging angular field. In a case where sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change an actual traveling direction of the hull to an angle at which the sunlight is outside the reference angular field.

[0009] The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] FIG. **1** is a schematic side diagram of a marine vessel to which a marine vessel steering control device is applied.
[0011] FIG. **2** is a block diagram of the marine vessel.
[0012] FIG. **3** is a conceptual diagram illustrating an imaging angular field and a reference angular field of two cameras.
[0013] FIG. **4** is a flowchart of an automated navigation control process.
[0014] FIG. **5** is a schematic diagram illustrating a relationship between the imaging angular field and an estimated location of sunlight.
[0015] FIGS. **6**A to **6**B are schematic diagrams illustrating a first example of a backlight avoidance process.
[0016] FIG. **7** is a schematic diagram illustrating a second example of the backlight avoidance process.
[0017] FIG. **8** is a schematic diagram illustrating a first example of a backlight preventive process.
[0018] FIG. **9** is a schematic diagram illustrating a second example of the backlight preventive process.
[0019] FIG. **10** is a diagram for illustrating a first example of a path correction process.
[0020] FIG. **11** is a diagram for illustrating a second example of the path correction process.
[0021] FIG. **12** is a diagram for illustrating a third example of the path correction process.
[0022] FIG. **13** is a diagram for illustrating a fourth example of the path correction process.
DETAILED DESCRIPTION OF THE EXAMPLE EMBODIMENTS
[0023] Hereinafter, example embodiments of the present invention will be described with reference to the drawings.
[0024] FIG. **1** is a schematic side diagram of a marine vessel to which a marine vessel steering control device according to an example embodiment of the present invention is applied. A marine vessel **100** includes a hull **101** and a marine propulsion device **102** mounted on the hull **101**. The marine propulsion device **102** is, for example, an outboard motor. Two or more marine propulsion devices **102** may be provided. The marine vessel **100** is communicatively connected to a server **40** via a network N.
[0025] The marine propulsion device **102** is attached to the hull **101** via an attachment unit **114**. The marine propulsion device **102** includes a drive source (for example, an engine) **103**. The drive source **103** may be an electric motor. The marine propulsion device **102** obtains thrust to move the hull **101** by a propeller rotated by a drive force of the drive source **103**. The attachment unit **114** includes a swivel bracket, a clamp bracket, a steering shaft, and a tilt shaft.
[0026] The attachment unit **114** further includes a power trim and tilt mechanism (PTT mechanism). The PTT mechanism causes the marine propulsion device **102** to pivot around the tilt shaft. This enables an inclination angle (a trim angle or a tilt angle) of the marine propulsion device **102** to be changed with respect to the hull **101**. The marine propulsion device **102** can pivot around the steering shaft with respect to the swivel bracket. During manual navigation, the marine propulsion device **102** pivots left and right by operating a steering wheel such that the marine vessel **100** is steered.
[0027] A main camera **24** and a sub camera **25** are provided at a bow of the hull **101**. The main camera **24** and the sub camera **25** are a first imager and a second imager, respectively, which are fixed to the hull **101** and capture images of a subject. Each camera may be attached, for example, to a bow rail (not illustrated) directly or via a support.
[0028] FIG. **2** is a block diagram of the marine vessel.
[0029] The marine vessel **100** includes, in addition to the main camera **24** and the sub camera **25**

described above, a navigation unit **21**, a sensor group **22**, a GNSS receiver **23**, and a display unit **26**. The marine vessel **100** also includes a CPU **11**, a ROM **12**, a RAM **13**, a memory **14**, a communication interface (I/F) **15**, a timer **16**, and an input unit **17**.

[0030] The CPU **11** controls the overall marine vessel **100**. The ROM **12** or the memory **14** stores a control program. The memory **14** also stores nautical chart information acquired from the server **40**. The CPU **11** implements various control processes by loading, in the RAM **13**, a control program stored in the ROM **12** or the like and executing the control program. The RAM **13** provides a work area when the CPU **11** executes the control program. The timer **16** measures time.

[0031] The navigation unit **21** includes, in addition to the drive source **103** (FIG. **1**), a steering, a remote controller, a throttle, a shift mechanism, a steering mechanism, and the like, and an element necessary for navigation. The sensor group **22** may include, in addition to a sensor to detect operation of the input unit **17**, an orientation sensor, an acceleration sensor, a speed sensor, an angular velocity sensor, an engine rpm sensor, a shift location sensor, a millimeter wave radar, and the like. A detection result obtained by the sensor group **22** is sent to the CPU **11**.

[0032] The GNSS receiver **23** periodically receives a global navigation satellite systems (GNSS) signal from a GNSS satellite. This enables the CPU **11** to acquire a current location of the marine vessel **100**.

[0033] The input unit **17** receives inputs of various settings, modes, or the like from a vessel operator of the marine vessel **100**. The communication I/F **15** can communicate with the server **40** via the network N, and can also communicate with an ECU that controls the drive source **103** of the navigation unit **21** via a Controller Area Network (CAN) or the like.

[0034] Imaging directions of the main camera **24** and the sub camera **25** are both set to a front side or a substantially front side. The display unit **26** displays various items of information. Captured images obtained by the cameras **24** and **25** are sent to the CPU **11** and displayed on the display unit **26** in an aspect according to the set mode. The CPU **11** configured or programmed to include and functions as a vessel operation controller mainly performs automated steering of the hull **101** based on the obtained captured images. A known method can be used as a basic method for achieving unmanned automated navigation. Thus, the automated navigation can be achieved by using, in addition to the captured images, detection information obtained from the sensor group **22**, location information obtained from the GNSS receiver **23**, or the like.

[0035] FIG. **3** is a conceptual diagram illustrating an imaging angular field and a reference angular field of each camera.

[0036] An imaging angular field **62** of the sub camera **25** is larger than an imaging angular field **61** of the main camera **24**. Since imaging directions of the cameras **24** and **25** are in common, angular field center-lines CL in a leftward-rightward direction are in common, and angular field center-lines in a vertical direction (not illustrated) are also in common. Thus, the imaging angular field **61** is inside the imaging angular field **62**.

[0037] A reference angular field **60** is an angular field used to determine whether or not a so-called backlit condition is observed. The reference angular field **60** is set as an angular field within the imaging angular field **61**. In the present example embodiment, the reference angular field **60** is smaller than the imaging angular field **61** and inside the imaging angular field **61**. The reference angular field **60** may be the same as the imaging angular field **61**.

[0038] In FIG. **3**, sunlight **50** represents a very bright light source such as the sun in the reference or imaging angular fields. The sunlight **50-1** is the sunlight **50** near the periphery of the reference angular field **60**, the sunlight **50-2** is the sunlight **50** near the periphery of the imaging angular field **61**, and the sunlight **50-3** is the sunlight **50** near the periphery of the imaging angular field **62**.

[0039] For example, the sunlight **50-1** and the sunlight **50-2** are contained in both captured images obtained by both the cameras **24** and **25**. However, the sunlight **50-3** is contained in the captured image obtained by the sub camera **25**, but is not contained in the captured image obtained by the main camera **24**.

[0040] FIG. **4** is a flowchart of an automated navigation control process. This process is achieved by causing the CPU **11** to load, in the RAM **13**, a program stored in the ROM **12** or the like and executing the program. This process is started in response to reception of an automated navigation start instruction, and is ended by an end instruction or unscheduled operation. If an image to be used for automated navigation is acquired in a backlit condition, quality of the image deteriorates, and it is difficult to perform appropriate automated vessel operations. In this respect, a backlight determination process and a vessel operation control depending on a determination result are executed.
[0041] In step S**101**, the CPU **11** configured or programmed to include and function as a determiner executes the backlight determination process. In the backlight determination process, it is determined whether or not a current imaging condition corresponds to the “backlit condition” or an “imminent backlit condition”.
[0042] A case where it is estimated that the sunlight **50** is reflected or the sunlight **50** is located within the reference angular field **60** is determined as the backlit condition (described with reference to FIGS. **3** and **5**).
[0043] A case where the sunlight **50** enters a region (frame region) within the imaging angular field **61** and outside the reference angular field **60** is determined as the imminent backlit condition (to be described with reference to FIG. **8**). A case where the sunlight **50** located outside the reference angular field **60** is predicted to enter the reference angular field **60** therefrom is also determined as the imminent backlit condition (to be described with reference to FIG. **9**). The imminent backlit condition may be defined as a critical condition close to the backlit condition.
[0044] A technique of determining the backlit condition will be described with reference to FIGS. **3** and **5**. FIG. **5** is a schematic diagram illustrating a relationship between the imaging angular field and an estimated location of sunlight. First, the CPU **11** determines, as the backlit condition, a case where the sunlight **50** is located within the reference angular field **60**. FIG. **3** illustrates a case where it is determined that the sunlight **50-1** is present within the reference angular field **60**. At that time, the backlit condition may be determined using one image obtained by the camera **24** or **25**, or both images obtained by the cameras **24** and **25**.
[0045] Specifically, the CPU **11** determines that the sunlight **50** is located within the reference angular field **60** in a case where the captured image obtained by the main camera **24** has a position or spot having a luminance equal to or larger than a threshold value within the reference angular field **60**. Alternatively, the CPU **11** determines that the sunlight **50** is located within the reference angular field **60** in a case where the captured image obtained by the sub camera **25** has a position or spot having a luminance equal to or larger than the threshold value within an angular field corresponding to the reference angular field **60**.
[0046] Alternatively, the backlit condition may be determined based on both of the estimated location of sunlight and a total light quantity of the image instead of a locally high luminance within the reference angular field **60**.
[0047] For example, as illustrated in FIG. **5**, the CPU **11** estimates a location of sunlight with respect to the reference angular field **60** based on a current location (latitude, longitude) of the hull **101**, an orientation of the main camera **24** and/or an orientation of the sub camera **25**, and a current time. The estimated current location of the sun is defined as an estimated location **51**. In the example of FIG. **5**, the estimated location **51** is observed within the reference angular field **60**. Then, the CPU **11** determines, as the backlit condition, a case where the estimated location **51** is observed within the reference angular field **60**, and the total light quantity of the image within the reference angular field **60** is equal to or larger than a predetermined light quantity. Thus, even if it is estimated that the estimated location **51** is present within the reference angular field **60**, this is not determined as the backlit condition when the total light quantity of the image within the reference angular field **60** is smaller than the predetermined light quantity due to rainy or cloudy weather. This is because an image without an exceptionally bright light source can be used for

automated navigation without any trouble.
[0048] In step S**102**, the CPU **11** determines whether or not a result of the backlight determination process in step S**101** corresponds to the backlit condition. Then, the CPU **11** proceeds to step S**103** in a case where the result of the backlight determination process corresponds to the backlit condition, and proceeds to step S**104** in a case where the result of the backlight determination process does not correspond to the backlit condition.
[0049] In step S**103**, since it is determined that the sunlight **50** is located within the reference angular field **60**, the CPU **11** executes a backlight avoidance process. In the backlight avoidance process, the CPU **11** configured or programmed to include and function as a vessel operation controller controls the navigation unit **21** to change the actual traveling direction of the hull **101** so that the sunlight **50** is outside the reference angular field **60**. For example, the CPU **11** calculates a thrust suitable to change the actual traveling direction and a direction thereof for each of the marine propulsion devices **102**, and controls each of the marine propulsion devices **102** depending on the calculation result. An example of the backlight avoidance process will be described with reference to FIGS. **6**A to **7**.
[0050] FIGS. **6**A and **6**B are schematic diagrams illustrating a first example of the backlight avoidance processing. In FIG. **6**A, similarly to FIG. **3**, the sunlight **50** is illustrated within the reference angular field **60** at the beginning. FIG. **6**B is a conceptual diagram illustrating the reference angular field **60** from above. An angle θ**0** of the reference angular field **60** in a leftward-rightward direction is 60°, for example. AC represents the actual traveling direction, and coincides with the angular field center-line CL on the captured image.
[0051] As illustrated in FIGS. **6**A and **6**B, in a case where the sunlight **50** is located within the reference angular field **60**, the CPU **11** changes the actual traveling direction AC of the hull **101** in one of a leftward direction or a rightward direction so that the angular field center-line CL of the reference angular field **60** becomes farther away from the sunlight **50**. In FIGS. **6**A and **6**B, since the sunlight **50** is located to the right of the angular field center-line CL, the CPU **11** changes the actual traveling direction AC to the left. When the actual traveling direction AC is changed, the amount of angle to be changed is half the angle (θ**0**/2) of the reference angular field. For example, if the reference angle is 60°, the amount of angle to be changed is 30°. By changing the actual traveling direction AC to the left by θ**0**/2, the sunlight **50** is outside the reference angular field **60**.
[0052] FIG. **7** is a schematic diagram illustrating a second example of the backlight avoidance process. In the example illustrated in FIG. **7**, the sunlight **50** is located within the reference angular field **60**. The CPU **11** changes the actual traveling direction AC of the hull **101** by at least a minimum turning angle θA in one of a leftward direction or a rightward direction in which a minimum turning angle θA in which the sunlight **50** is outside the reference angular field **60** becomes small. The minimum turning angle θA is calculated by the CPU **11**. In the example illustrated in FIG. **7**, the sunlight **50** is located to the right of the angular field center-line CL. In this case, the CPU **11** changes the actual traveling direction AC to the left by an angle equal to or larger than the minimum turning angle θA. This causes the sunlight **50** to be outside the reference angular field **60**.
[0053] In the first example and the second example, in a case where the sunlight **50** is observed on the angular field center-line CL, the actual traveling direction AC may be changed to a predetermined direction of a leftward direction or a rightward direction. Either process of the first example or the second example may be used as the backlight avoidance process. A user may designate a process to be activated in advance. After step S**103**, the CPU **11** returns to step S**101**.
[0054] In step S**104**, the CPU **11** determines whether or not the result of the backlight determination process (step S**101**) corresponds to the imminent backlit condition. Then, the CPU **11** proceeds to step S**105** in a case where the result of the backlight determination process corresponds to the imminent backlit condition, and proceeds to step S**106** in a case where the result of the backlight determination process does not correspond to the imminent backlit condition.

[0055] In step S**105**, the CPU **11** executes a backlight preventive process. The backlight preventive process is a process in which the CPU **11** configured or programmed to include and function as the vessel operation controller controls the navigation unit **21** to change the actual traveling direction AC of the hull **101** by a first predetermined angle θ**1** in one of a leftward direction or a rightward direction in which the angular field center-line CL (the center) of the reference angular field **60** becomes farther away from the sunlight **50**. Examples of the imminent backlit condition and the backlight preventive process will be described with reference to FIGS. **8** and **9**.
[0056] FIG. **8** is a schematic diagram illustrating a first example of the backlight preventive process. In the example illustrated in FIG. **8**, the sunlight **50** is observed in a region (a frame region) within the imaging angular field **61** and outside the reference angular field **60**. Since the sunlight **50** is observed outside the reference angular field **60**, an imaging condition does not reach the backlit condition, but there is a possibility that the imaging condition will soon reach the backlit condition. In this respect, the CPU **11** changes the actual traveling direction AC by the first predetermined angle θ**1** to the left side which is one of a leftward direction or a rightward direction in which the angular field center-line CL becomes farther away from the sunlight **50**. This makes it possible to avoid the backlit condition in advance.
[0057] FIG. **9** is a schematic diagram illustrating a second example of the backlight preventive process. In the example illustrated in FIG. **9**, the CPU **11** acquires the relative estimated location **51** of the sunlight with respect to the reference angular field **60** based on the current location (latitude, longitude) of the hull **101**, an orientation of the main camera **24** or an orientation of the sub camera **25**, and a current time. The CPU **11** constantly acquires weather information via the network N. The CPU **11** predicts a possibility that the sunlight **50** will enter the reference angular field **60** based on the estimated location **51** and the weather information. Then, in a case where the sunlight **50** is predicted to enter the reference angular field **60**, the CPU **11** changes the actual traveling direction AC of the hull **101** by the first predetermined angle θ**1** in one of a leftward direction or a rightward direction in which the angular field center-line CL (the center) of the reference angular field **60** becomes farther away from the estimated location **51** (the sunlight **50**).
[0058] In the example illustrated in FIG. **9**, it is assumed that the estimated location **51** is observed on the right of the reference angular field **60**, and the estimated location **51** is predicted to move relatively to the left. At this time, based on the weather information, in a case where the weather is sunny or the like and there is a possibility that the sun will be reflected in the captured image, it is predicted that there is a possibility that the sunlight **50** will soon enter the reference angular field **60** and cause the backlit condition. In this respect, the CPU **11** changes the actual traveling direction AC by the first predetermined angle θ**1** to the left side which is one of a leftward direction or a rightward direction in which the angular field center-line CL becomes farther away from the sunlight **50**. This makes it possible to avoid the backlit condition in advance.
[0059] Based on the weather information, in a case where the weather is rainy or the like, the backlit condition is not caused even if the estimated location **51** is located within the reference angular field **60**. Thus, in step S**104** described above, it is not determined as the imminent backlit condition.
[0060] The first predetermined angle θ**1** may be, for example, 10°. In the first example and the second example, the first predetermined angles θ**1** may have the same value or different values. After step S**105**, the CPU **11** returns to step S**101**.
[0061] In step S**106**, the CPU **11** calculates displacement OB between a target traveling direction TC and an actual traveling direction AC of the hull **101**, and determines whether or not there is path displacement. In a case where the displacement OB is larger than a predetermined displacement amount, it is determined that the path displacement is caused between the target traveling direction TC and the actual traveling direction AC. The predetermined displacement amount is a value equal to or smaller than a second predetermined angle θ**2** and equal to or smaller than half the angle (θ**0**/2) of the reference angular field. A magnitude relationship between the second predetermined

angle θ**2** and θ**0**/2 is not limited. The value of the second predetermined angle θ**2** is not limited, and may be, for example, 12°. In the present example embodiment, the predetermined displacement amount is the same as the second predetermined angle θ**2**.

[0062] As a result of the determination in step S**106**, the CPU **11** proceeds to step S**107** in a case where there is path displacement, and returns to step S**101** in a case where there is no path displacement. Steps S**106** and S**107** are executed generally in a case where the sunlight **50** is not located within the reference angular field **60**.

[0063] In step S**107**, the CPU **11** executes a path correction process, and then returns to step S**101**. The path correction process is a process in which the CPU **11** configured or programmed to include and function as the vessel operation controller controls the navigation unit **21** to reduce or eliminate the path displacement to zero. With reference to FIGS. **10** to **13**, four techniques of the path correction process (first to fourth examples) will be described. One of these four techniques may be applied, or two or more techniques may be used in a combination as long as the techniques are not incompatible.

[0064] FIGS. **10**, **11**, **12**, and **13** are diagrams for illustrating a first example, a second example, a third example, and a fourth example of the path correction process, respectively. As a result of the path correction process, the sunlight **50** may be located within the reference angular field **60**. However, by returning to step S**101**, the backlight determination process is executed again, and then the backlight avoidance process or the like is executed as necessary.

[0065] In the first example illustrated in FIG. **10**, the displacement OB larger than half the angle of the reference angular field **60** in the leftward-rightward direction is caused ("θ**0**/2"<θB). In this case, the CPU **11** changes the actual traveling direction AC by the angle (θ**0**) of the reference angular field in one of a leftward direction or a rightward direction which includes the target traveling direction TC. In the example illustrated in FIG. **10**, the CPU **11** changes the actual traveling direction AC to the right side by the angle (θ**0**) of the reference angular field. By controlling in this manner, the displacement OB becomes small. This can prevent a state of being displaced in one direction from being maintained for a long time as long as the sunlight **50** is not located within reference angular field **60**.

[0066] In the second example illustrated in FIG. **11**, the displacement OB larger than the second predetermined angle θ**2** is caused in the leftward-rightward direction (θ**2**<θB). In this case, the CPU **11** changes the actual traveling direction AC by a third predetermined angle θ**3** in a direction approaching the target traveling direction TC. The third predetermined angle θ**3** is a value that is not larger than the second predetermined angle θ**2**, and may be, for example, 10°. As a result, the displacement OB is decreased. This enables the actual traveling direction AC to be gradually returned to the target traveling direction TC as long as the sunlight **50** is not located within the reference angular field **60**.

[0067] Also in the third example illustrated in FIG. **12**, the displacement OB larger than the second predetermined angle θ**2** is caused in the leftward-rightward direction (θ**2**<θB). In this case, when the total light quantity of the captured image obtained by the sub camera **25** becomes smaller than the predetermined light quantity, the CPU **11** returns the actual traveling direction AC to the target traveling direction TC after waiting an elapsed predetermined period of time. It is assumed that the total light quantity becomes smaller than the predetermined light quantity due to the sunset or the rainy or cloudy weather. In this case, even in a case where the estimated location **51** is found within the reference angular field **60**, it is predicted that the backlit condition is not caused. Since the necessity of the backlight avoidance is reduced, the displacement amount OB is set to zero by returning the actual traveling direction AC to the target traveling direction TC at once. A reason for waiting for the elapse of the predetermined period of time is to prevent divergence of control by considering a case where the total light quantity temporarily becomes smaller than the predetermined light quantity. It is not essential to wait for the elapsed predetermined period of time.

[0068] In the fourth example illustrated in FIG. **13**, the CPU **11** acquires the relative estimated

location **51** of the sunlight with respect to the reference angular field **60** based on the current location (latitude, longitude) of the hull **101**, the orientation of the main camera **24** or the orientation of the sub camera **25**, and the current time. In the example illustrated in FIG. **13**, the target traveling direction TC is closer to the estimated location **51** than the angular field center-line CL of the reference angular field **60** is. Moreover, it is assumed that the estimated location **51** (the sunlight **50**) is predicted to become gradually farther away from the reference angular field **60** outside the reference angular field **60** from the current orientation, location, time, or the like.
[0069] In such a case, whenever the estimated location **51** becomes farther away by a fourth predetermined angle θ**4** (for example, 10°), the CPU **11** changes the actual traveling direction AC by the fourth predetermined angle θ**4** in one of a leftward direction or a rightward direction which approaches the estimated location **51** (the right direction in the example illustrated in FIG. **13**). This enables the actual traveling direction AC to gradually approach the target traveling direction TC in a situation in which the sunlight **50** moves away, in a case where a certain direction of a leftward direction or a rightward direction in which the sunlight **50** is observed is the same as a displacement direction of the target traveling direction TC.
[0070] According to the present example embodiment, in a case where the sunlight **50** is located within the reference angular field **60**, the CPU **11** changes the actual traveling direction to an angle at which the sunlight **50** moves away from the reference angular field **60**. This curbs acquisition of an image to be used for automated navigation in the backlit condition or the like. Accordingly, deterioration in quality of an acquired image due to the sunlight can be reduced, and the appropriate automated vessel operation can be achieved.
[0071] The angles θ**0**, θ**1**, θ**2**, θ**3**, θ**4**, and θA are stored in the memory **14** in advance, but may be changed afterwards. The values of the angles θ**0**, θ**1**, θ**2**, θ**3**, θ**4**, and θA are not limited to the values shown as examples.
[0072] It is not essential to execute the backlight preventive process (S**105**) or the path correction process (S**107**). Even if a mode of executing only the backlight preventive process (S**105**) or the path correction process (S**107**) without executing the backlight avoidance process (S**103**) is executed, this contributes to reduction of deterioration in quality of the acquired image.
[0073] Although the present invention has been described in detail based on the example embodiments thereof, the present invention is not limited to these specific example embodiments, and various example embodiments in a range without departing from the gist of the present invention are also included in the present invention.
[0074] Example embodiments of the present invention are also applicable to jet boats, in addition to various marine vessels propelled by outboard motors, inboard motors, or inboard/outboard motors.
[0075] Example embodiments of the present invention can also be achieved by processes in which a program for executing one or more functions of the above-described example embodiments is supplied to a system or a device via a network or a non-transitory storage medium, and one or more processors of a computer of the system or the device reads and executes the program. The above program and a non-transitory storage medium storing the above program are also example embodiments of the present invention. Example embodiments of the present invention can also be achieved by circuits (for example, an ASIC) that provide one or more functions.
[0076] While example embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A marine vessel steering control device comprising: a first imager fixed to a hull to capture an image of a subject; and a controller configured or programmed to include: a vessel operation controller configured or programmed to perform an automated vessel operation of the hull based on a captured image obtained by the first imager; and a determiner configured or programmed to determine, based on an imaging angular field of the first imager, whether or not sunlight is located within a reference angular field defined as an angular field within the imaging angular field; wherein in a case where the sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change an actual traveling direction of the hull to an angle at which the sunlight is located outside the reference angular field.
2. The marine vessel steering control device according to claim 1, wherein the determiner is configured or programmed to determine that the sunlight is located within the reference angular field in a case where a position having a luminance equal to or higher than a threshold value is located within the reference angular field.
3. The marine vessel steering control device according to claim 1, further comprising: a second imager including a wide imaging angular field including the imaging angular field of the first imager; wherein the determiner is configured or programmed to determine that the sunlight is located within the reference angular field in a case where a captured image obtained by the second imager including a position having a luminance equal to or higher than a threshold value within an angular field corresponding to the reference angular field.
4. The marine vessel steering control device according to claim 1, wherein the determiner is configured or programmed to estimate a location of the sunlight based on a location of the hull and a current time, and to determine whether or not the sunlight is located within the reference angular field based on the estimated location of the sunlight and a total light quantity of an image within the reference angular field.
5. The marine vessel steering control device according to claim 4, wherein the determiner is configured or programmed to determine that the sunlight is located within the reference angular field in a case where the estimated location of the sunlight is within the reference angular field and the total light quantity of the image within the reference angular field is equal to or larger than a predetermined light quantity.
6. The marine vessel steering control device according to claim 1, wherein, in a case where the sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by half an angle of the reference angular field in one of a leftward direction or a rightward direction in which a center of the reference angular field becomes farther away from the sunlight.
7. The marine vessel steering control device according to claim 1, wherein, in a case where the sunlight is located within the reference angular field, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by at least a minimum turning angle in one of a leftward direction or a rightward direction in which the minimum turning angle in which the sunlight is located outside the reference angular field becomes smaller.
8. The marine vessel steering control device according to claim 1, wherein the reference angular field is smaller than the imaging angular field of the first imager; and in a case where the sunlight enters a region within the imaging angular field and outside the reference angular field, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by a first predetermined angle in one of a leftward direction or a rightward direction in which a center of the reference angular field becomes farther away from the sunlight.
9. The marine vessel steering control device according to claim 1, wherein the determiner is configured or programmed to estimate a location of the sunlight based on a location of the hull and a current time; and the vessel operation controller is configured or programmed to predict whether or not the sunlight will enter the reference angular field based on the estimated location of the

sunlight and weather information, and in a case where the vessel operation controller predicts that the sunlight will enter the reference angular field, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by a first predetermined angle in one of a leftward direction or a rightward direction in which a center of the reference angular field becomes farther away from the sunlight.

10. The marine vessel steering control device according to claim 1, wherein, in a case where the sunlight is not located within the reference angular field and a displacement larger than half an angle of the reference angular field is caused between a target traveling direction and the actual traveling direction of the hull, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by the angle of the reference angular field in one of a leftward direction or a rightward direction which includes the target traveling direction.

11. The marine vessel steering control device according to claim 1, wherein, in a case where the sunlight is not located within the reference angular field and a displacement larger than a second predetermined angle is caused between a target traveling direction and the actual traveling direction of the hull, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by a third predetermined angle that is not larger than the second predetermined angle in a direction approaching the target traveling direction.

12. The marine vessel steering control device according to claim 1, further comprising: a second imager having an imaging angular field including the imaging angular field of the first imager; wherein in a case where the sunlight is not located within the reference angular field and a displacement larger than a second predetermined angle is caused between a target traveling direction and the actual traveling direction of the hull, when a total light quantity of a captured image obtained by the second imager becomes smaller than a predetermined light quantity, the vessel operation controller is configured or programmed to return the actual traveling direction of the hull to the target traveling direction after waiting an elapsed predetermined period of time.

13. The marine vessel steering control device according to claim 1, wherein the determiner is configured or programmed to estimate a location of the sunlight based on a location of the hull and a current time; and in a case where the sunlight is predicted to move away from the reference angular field outside the reference angular field and a target traveling direction of the hull is closer to the sunlight than a center of the reference angular field, the vessel operation controller is configured or programmed to change the actual traveling direction of the hull by a fourth predetermined angle in one of a leftward direction or a rightward direction in which the actual traveling direction approaches the sunlight whenever the location of the sunlight is farther away from the center of the reference angular field by the fourth predetermined angle.

14. A marine vessel steering control method for a marine vessel including a hull to which an imager that captures an image of a subject is fixed, the method comprising: performing an automated vessel operation of the hull based on a captured image obtained by the imager; determining, based on an imaging angular field of the imager, whether or not sunlight is located within a reference angular field defined as an angular field within the imaging angular field; and changing an actual traveling direction of the hull to an angle at which the sunlight is located outside the reference angular field in a case where the sunlight is located within the reference angular field.

15. A marine vessel comprising: the marine vessel steering control device according to claim 1.